

Luiz Sérgio da Silva Filho

SOFTWARE ENGINEER | FULL STACK DEVELOPER

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Open to hybrid, remote or on-site roles

SUMMARY

Full stack software engineer with ten systems of my own delivered end to end, from architecture to deployment, across domains such as labor-court expert work, school management, automotive marketplaces and personal finance, one of them already running in production with real customers. The edge is in the rigor behind delivery: money as Decimal and never a float, balances derived from an append-only ledger, multi-tenant isolation with Row-Level Security verified by explicit tests, idempotent schedulers and webhooks, and AI layers that never write to the database without application-side validation. Bachelor's in Information Technology, with a background in IT support, running my own business and pedagogical coordination, experience that keeps technical decisions close to the real pain of the people using the system.

PROFESSIONAL EXPERIENCE

Pedagogical Coordinator

Feb 2026 – Present

Children's Programming and Robotics School · Barretos, SP, Brazil

- Built the pedagogical coordination from the ground up when the course opened: organized the classes, defined the curriculum by age group and assembled the teaching team, serving students from 7 to 18 years old.
- Age-progressive curriculum: younger classes work with robotics (Arduino, micro:bit) and block-based programming; teenage classes start with Python and move on to Git, HTML, CSS, JavaScript, Tailwind CSS, daisyUI, React, Node.js, MongoDB, JWT and Swagger as their full stack foundation.
- Teach Python to the teenage classes, covering programming logic, basic data structures and small hands-on projects.
- Faced with the limitations of the system the unit was using for pedagogical management and tracking, I ran the entire lifecycle of a replacement system on my own: gathered the requirements from my own coordination routine, designed the data model and the role-based permissions (director, coordinator, teacher), and validated every release against real school usage before putting it into production. The product is Atlas Educação, detailed below.

Owner and Partner

Jun 2020 – Present

Empório do Bebê · Barretos, SP, Brazil

- Responsible for running the business: point of sale and inventory control, online sales channel and digital catalog, and automation of administrative processes through spreadsheets. The systems built in parallel are detailed in the next section.

IT Support Analyst

Aug 2013 – Dec 2019

Sesi São Paulo · Limeira and Barretos, SP, Brazil

- IT support for staff, teachers and students: helpdesk ticket handling, administrative and educational systems, and support for classroom technology.
- Technical mentoring for educational robotics teams: support with programming and with planning the execution order of the robot missions to solve the proposed challenges, in competitions such as FLL (First Lego League) and OBR (Brazilian Robotics Olympiad).

INDEPENDENT TECHNICAL PROJECTS

Complete systems conceived, architected and shipped independently, alongside my work as a partner at Empório do Bebê. Technical breakdown of every project at luizsergio.vercel.app.

- **PeritoCalc** · SaaS for court-appointed labor experts. The extraction engine routes every field through eight cost-ordered layers, from native PDF text to computer vision as a last resort, recording each decision in a per-layer auditable ledger. The .PJC file generator runs in pure Python 3 with zero external dependencies and produces the compressed XML ready for direct import into PJe-Calc. Next.js 15, Supabase with Row-Level Security, Stripe.
- **Sondei** · Automotive marketplace in a monorepo, with the full FIPE price table (11,354 versions) and IBGE geography (5,571 municipalities) imported, backed by 443 integration tests running against a real Postgres, including explicit cross-dealership isolation checks. The recommendation advisor never gets a database connection: it only extracts structured filters from the question, and the answer is assembled on top of an application-controlled query, making it impossible to suggest a vehicle that is not in stock. Turborepo, Next.js, Drizzle ORM, BullMQ.
- **Meu Assessor** · Personal finance management over WhatsApp and a web dashboard, with a domain core in plain TypeScript following DDD, free of framework coupling. Money always as Decimal and never a float, with exact cent distribution across installments, and recurrence schedulers idempotent by their own key. Argon2id, TOTP-based MFA and personal data export and deletion routines for data-protection compliance. NestJS, Prisma, BullMQ.
- **Liquida Sorte (in production)** · Inventory liquidation organized into lots and cards, with auditable draws and cryptographically generated random numbers. Customer balance is always derived from an append-only ledger, never edited directly, and every payment webhook is idempotent. Because the product is legally sensitive, public checkout and real draw execution sit behind feature flags that are off by default. Already serving real customers.
- **Atlas Educação** · Multi-school pedagogical management portal, with a dropout-risk engine whose attendance, consecutive-absence and grade-drop thresholds are configurable per school. A seven-provider LLM chain absorbs rate limits applied per organization, personal data is anonymized before anything is sent, and the product stays fully functional with no AI key configured at all.
- **Ponto Eletrônico** · Multi-site workforce time tracking with facial recognition running in an isolated Python microservice (FastAPI, dlib and face_recognition) and a similarity threshold configurable through an environment variable. Covers standard and 12x36 shift patterns, break tolerances, absence-justification approval and attendance reports. Next.js 14, Prisma, PostgreSQL and Docker Compose.
- **Canal Dark** · Electron desktop application that drives the full short-form video production pipeline: script, per-scene imagery, narration and final assembly with a bundled FFmpeg. Every stage has a free provider that works with no API key and automatically takes over if a paid provider fails. Credentials are encrypted in the operating system's native vault.
- **Other systems in the same line:** retail ERP in NestJS and Fastify on a Turborepo monorepo, with nine-role RBAC, per-action audit trail and a pluggable tax layer for electronic invoicing; a car dealership platform in Next.js 16 with monetary values as integer cents and accent-insensitive search; and WhatsApp customer service with semantic search over the catalog and handoff to a human agent.

EDUCATION

Postgraduate Degree in Higher Education Teaching

2015

Faculdade XV de Agosto

Bachelor's Degree in Information Technology

2008 – 2011

Faculdade Barretos

TECHNICAL SKILLS

Frontend & languages: TypeScript, Python, React.js, Next.js

Backend & APIs: Node.js, NestJS, REST API, Webhooks

Data & architecture: PostgreSQL, Prisma ORM, Drizzle ORM, Row-Level Security, Multi-tenant Architecture, Redis

Artificial intelligence: Anthropic Claude API, OpenAI API, LLM Integration, Prompt Engineering, AI-Assisted Development

Infrastructure & engineering practices: Docker, CI/CD, Git, BullMQ, Automated software testing, Stripe API, Process automation, n8n